

Work Power Energy

Questions with Answer Keys

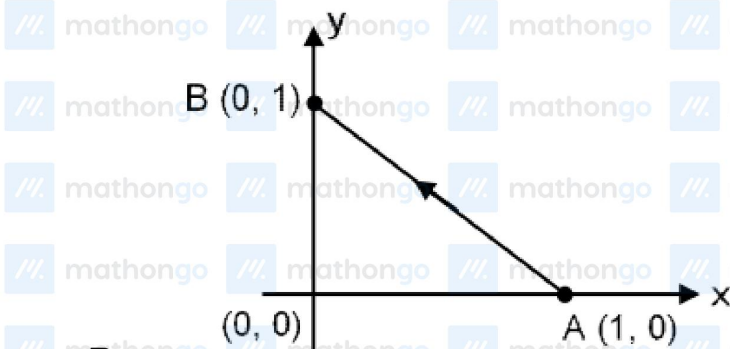
JEE Main 2020 Chapterwise

Physics

Q1 JEE Main 2020 - 9 January (Morning)

Particle moves from point A to point B along the line shown in figure under the action of force.

$\vec{F} = -x\hat{i} + y\hat{j}$. Determine the work done on the particle by $\vec{F}$ in moving the particle from point A to

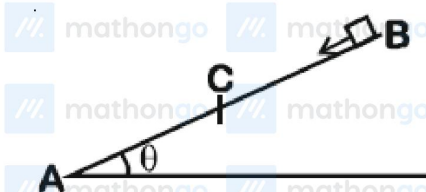


point B .

- (A) $1 J$
- (B) $\frac{1}{2} J$
- (C) $2 J$
- (D) $3 J$

Q2 JEE Main 2020 - 2 September (Morning)

A small block starts slipping down from a point B on an inclined plane AB , which is making an



angle θ with the horizontal section BC is smooth and the remaining section CA is rough with a coefficient of friction μ . It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If $BC = 2AC$, the coefficient of friction is given by $\mu = k \tan \theta$. The value of k is

Q3 JEE Main 2020 - 3 September (Evening)

A block starts moving up an inclined plane of inclination 30° with an initial velocity of v_0 . It comes back to its initial position with velocity

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$\frac{v_0}{2}$. The value of the coefficient of kinetic friction between the block and the inclined plane is close to $\frac{1}{1000}$. The nearest integer to 1 is _____.

Q4 JEE Main 2020 - 4 September (Evening)

A person pushes a box on a rough horizontal platform surface. He applies a force of 200N over a distance of 15m. Thereafter, he gets progressively tired and his applied force reduces linearly with distance to 100N. The total distance through which the box has been moved is 30m. What is the work done by the person during the total movement of the box?

- (A) 5690J
- (B) 3280J
- (C) 5250J
- (D) 2780J

Q5 JEE Main 2020 - 5 September (Morning)

A beam of electrons of energy E scatters from a target having atomic spacing of $1\text{ \AA}$. The first maximum intensity occurs at $\theta = 60^\circ$. Then E (in eV) is (Planck constant $h = 6.64 \times 10^{-34}\text{ Js}$, $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$, electron mass $m = 9.1 \times 10^{-31}\text{ kg}$)

Q6 JEE Main 2020 - 5 September (Evening)

A body of mass 2kg is driven by an engine delivering a constant power of 1J/s. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m)

Q7 JEE Main 2020 - 6 September (Morning)

If the potential energy between two molecules is given by $U = -\frac{A}{r^6} + \frac{B}{r^{12}}$, then at equilibrium, separation between molecules, and the potential energy are

- (A) $\left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B}$
- (B) $\left(\frac{B}{2A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B}$

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(C) $\left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{4B}$

(D) $\left(\frac{B}{A}\right)^{\frac{1}{6}}, 0$

Q8 JEE Main 2020 - 7 January (Morning)

A 60 HP electric motor lifts an elevator having a maximum total load capacity of 2000 kg . If the frictional force on the elevator is 4000 N , the speed of the elevator at full load is close to : ($1 \text{ HP} = 746 \text{ W}$, $g = 10 \text{ m/s}^2$)

(A) 1.9 m/s

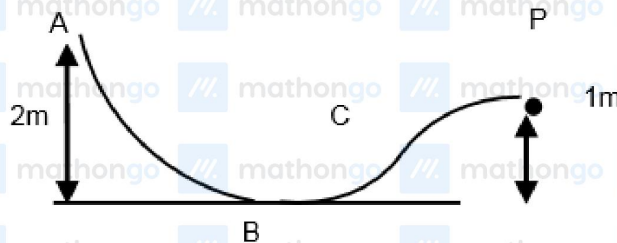
(B) 1.7 m/s

(C) 2 m/s

(D) 1.5 m/s

Q9 JEE Main 2020 - 7 January (Morning)

A particle ($m = 1 \text{ kg}$) slides down a frictionless track (AOC) starting from rest at a point A (height 2 m). After reaching C , the particle continues to move freely in air as a projectile. When it reaches its highest point P (height 1 m), the kinetic energy of the particle (in J) is : (Figure drawn is schematic and



not to scale; take $g = 10 \text{ m/s}^2$)

Q10 JEE Main 2020 - 7 January (Evening)

An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg . The mass of the elevator itself is 920 kg and it moves with a constant speed of 3 m/s . The frictional force opposing the motion is 6000 N . If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator ($g = 10 \text{ m/s}^2$) must be at least :

(A) 66000 W

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Physics

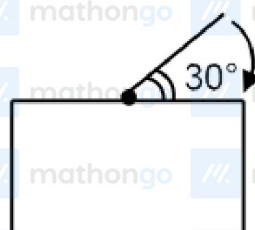
(B) 63248 W

(C) 48000 W

(D) 56320 W

Q11 JEE Main 2020 - 9 January (Morning)

One end of a straight uniform 1 m long bar is pivoted on horizontal table. It is released from rest when it makes an angle 30° from the horizontal (see figure). Its angular speed when it hits the table is given as



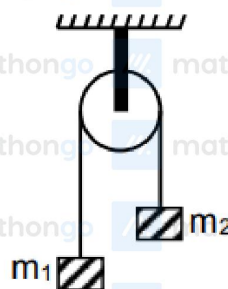
$\sqrt{n} \text{ s}^{-1}$, where n is an integer. The value of n is

Q12 JEE Main 2020 - 9 January (Morning)

A body of mass $m = 10\text{ kg}$ is attached to one end of a wire of length 0.3 m . The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station is (Breaking stress of wire $= 4.8 \times 10^7 \text{ Nm}^{-2}$ and area of cross section of the wire $= 10^{-2} \text{ cm}^2$) is

Q13 JEE Main 2020 - 9 January (Evening)

A uniformly thick wheel with moment of inertial I and radius R is free to rotate about its centre of mass (see figure). A mass-less string is wrapped over its rim and two blocks of masses m_1 and m_2 ($m_1 > m_2$) are attached to the ends of the string. The system is released from rest. The angular



speed of the wheel when m_1 descends by a distance h is :

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- (A) $\left[\frac{m_1+m_2}{(m_1+m_2)R^2+1} \right]^{\frac{1}{2}} gh$
- (B) $\left[\frac{2(m_1-m_2)gh}{(m_1+m_2)R^2+1} \right]^{\frac{1}{2}}$
- (C) $\left[\frac{2(m_1+m_2)gh}{(m_1+m_2)R^2+1} \right]^{\frac{1}{2}}$
- (D) $\left[\frac{(m_1+m_2)}{(m_1+m_2)R^2+1} \right]^{\frac{1}{2}} gh$

Q14 JEE Main 2020 - 9 January (Evening)

A spring mass system (mass m , spring constant k and natural length ℓ) rests in equilibrium on a horizontal disc. The free end of the spring is fixed at the centre of the disc. If the disc together with spring mass system, rotates about it's axis with an angular velocity ω , ($k \gg m\omega^2$) the relative change in the length of the spring is best given by the option :

- (A) $\sqrt{\frac{2}{3}} \left(\frac{m\omega^2}{k} \right)$
- (B) $\frac{m\omega^2}{3k}$
- (C) $\frac{2m\omega^2}{k}$
- (D) $\frac{m\omega^2}{k}$

Answer Key

Q1 (A)

Q2 (3)

Q3 (346)

Q4 (C)

Q5 (50.47)

Q6 (18)

Q7 (C)

Q8 (A)

Q9 (10)

Q10 (A)

Q11 (15)

Q12 (4)

Q13 (B)

Q14 (D)

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Hints & Solutions

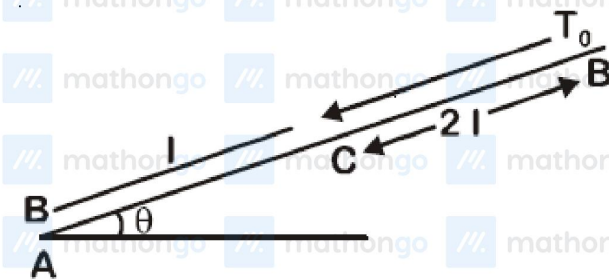
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Physics

Q1

$$\begin{aligned} W &= \int \vec{F} \cdot d\vec{s} \\ &= (-x\hat{i} + y\hat{j}) \cdot (dx\hat{i} + dy\hat{j}) \\ &= \int_0^1 -x dx + \int_0^1 y dy \\ &= -\frac{x^2}{2} \Big|_1^0 + \frac{y^2}{2} \Big|_0^1 = \left(0 + \frac{1}{2}\right) + \left(\frac{1}{2}\right) = 1 \text{ J} \end{aligned}$$

Q2



Work done by all the force/s is equal to zero.

$$\therefore 0 = mg3 \sin \theta - \mu mg \cos \theta = 0$$

$$3 \sin \theta = \mu \cos \theta \Rightarrow \mu = 3 \tan \theta$$

$$\therefore k = 3$$

Q3

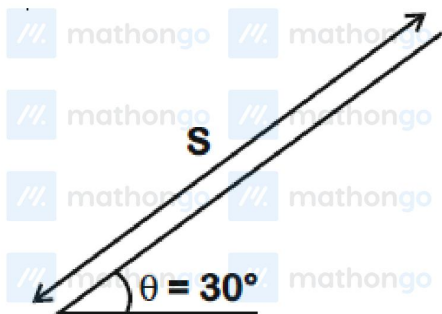
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Hints & Solutions

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Physics

For upward motion

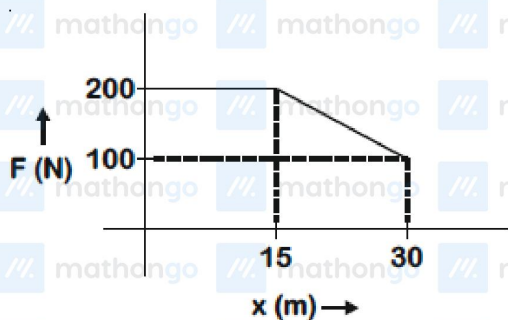


$$\frac{1}{2}mv_0^2 = \left(mg \frac{s}{2} + \mu mg \frac{\sqrt{3}}{2} s \right)$$
$$\frac{1}{2} \frac{mv_0^2}{4} = \left(mg \frac{s}{2} - \mu mg \frac{\sqrt{3}}{2} \cdot S \right)$$

for downward

$$4 = \frac{(1+\mu\sqrt{3})}{(1-\mu\sqrt{3})}$$
$$\Rightarrow 4 - 4\mu\sqrt{3} = 1 + \mu\sqrt{3}$$
$$\Rightarrow 3 = 5\mu\sqrt{3} \quad \therefore \mu = \frac{\sqrt{3}}{5} \approx \frac{346.41}{1000}$$
$$1 = 346$$

Q4



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Physics

Work = Area of F-x curve

$$= 200 \times 15 + \frac{1}{2}(200 + 100) \times 15$$
$$= 5250 \text{ J}$$

Q5

$$\frac{1}{2}mv^2 = E$$

$$\Rightarrow P = \sqrt{2Em} \quad \therefore \lambda = \frac{h}{\sqrt{2Em}}$$

For first maxima,

$$2d \sin \theta = \lambda$$

$$\Rightarrow 2 \times 10^{-10} \times \frac{\sqrt{3}}{2} = \frac{h}{\sqrt{2Em}}$$

$$\Rightarrow 2Em = \frac{(10^{10} \times h)^2}{3}$$

$$\Rightarrow E = \frac{(10^{10} \times h)^2}{6m_e \times e} \text{ (eV)}$$

$$\Rightarrow E = 50.47 \text{ eV}$$

Q6

$$pt = \frac{1}{2}mv^2$$

$$\Rightarrow v = \sqrt{\frac{2pt}{m}}$$

$$\Rightarrow \frac{ds}{dt} = \sqrt{\frac{2pt}{m}}$$

$$\Rightarrow \int_0^s ds = \int_0^t \sqrt{\frac{2pt}{m}} dt$$

$$\Rightarrow s = \left(\sqrt{\frac{2p}{m}} \right) \left(\frac{2t^{\frac{3}{2}}}{3} \right) = (1) \times (2) \times \frac{27}{3} = 18$$

Q7

$$F = -\frac{dU}{dr}$$

$$= -\left[\frac{6A}{r^7} - \frac{12B}{r^{13}} \right]$$

$$F = 0$$

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$$\Rightarrow r = \left(\frac{2B}{A} \right)^{1/6}$$

$$U \left(\text{at } r = \left(\frac{2B}{A} \right)^{1/6} \right) = -\frac{A^2}{4B}$$

Q8

$$4000 \times V + mg \times V = P$$

$$\frac{60 \times 746}{4000 + 20000} = V$$

$$V = 1.86 \text{ m/s} \approx 1.9 \text{ m/s}.$$

Q9

$$KE = PE_1 - PE_2 = mgh_1 - mgh_2$$

$$= 1 \times 10 \times 2 - 1 \times 10 \times 1 = 10 \text{ J}$$

Q10

Net force on motor will be

$$F_m = [920 + 68(10)]g + 6000$$
$$= 22000 \text{ N}$$

So, required power for motor

$$P_m = \vec{F}_m \cdot \vec{v}$$
$$= 22000 \times 3$$

$$= 66000 \text{ watt}$$

Q11

$$mg \frac{\ell}{2} \sin 30^\circ = \frac{1}{2} \frac{m\ell^2}{3} \omega^2$$

Solving

$$\omega^2 = 15$$

$$\omega = \sqrt{15}$$

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Physics

Q12

$$\frac{T}{A} = \sigma$$

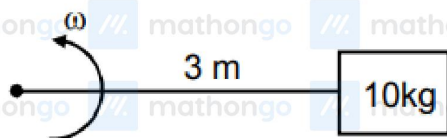
... (1)

$$T = m\omega^2 \ell$$

... (2)

Solving

$$\omega = 4 \text{ rad/s}$$



Q13

$$k_i + U_i = k_f + U_f$$

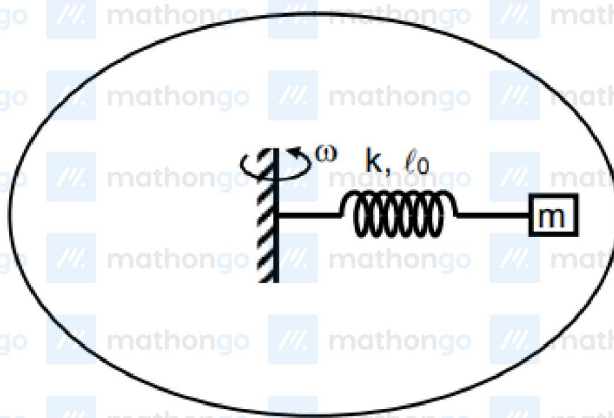
$$0 + 0 = \frac{1}{2} m_2 v^2 + \frac{1}{2} m_1 v^2 + \frac{1}{2} I \omega^2 - m_1 gh + m_2 gh$$

$$(m_1 + m_2) gh = \frac{1}{2} m_2 (\omega R)^2 + \frac{1}{2} m_1 (\omega R)^2 + \frac{1}{2} I \omega^2$$

$$\sqrt{\frac{2(m_1 - m_2) gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right) R^2}} = \omega$$

$$\frac{1}{R} \sqrt{\frac{2(m_1 - m_2) gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right)}}$$

Q14



$$m\omega^2(\ell_0 + x) = kx$$

$$\left(\frac{\ell_0}{x} + 1\right) = \frac{k}{m\omega^2}$$

$$x = \frac{\ell_0 m \omega^2}{k - m \omega^2}$$

$$k \gg m \omega^2$$

$$\text{So, } \frac{x}{\ell_0} \text{ is equal to } \frac{m \omega^2}{k}$$